deviceCare - Voltage Protector

STM32F103 Blue Pill: disconnects a load on overvoltage, undervoltage or overcurrent

What it does

Two analog inputs (voltage and current sensor) are sampled every 100 ms. If either leaves its limits, the relay on PB0 drops the load. Limits can be changed on the device or from a computer.

Protection logic

- Overvoltage: V > **250 V** (default)
- Undervoltage: V < **170 V**
- Overcurrent: I > **10 A**
- Scaling: V = ADC x 330/4095, I = ADC x 20/4095 (12-bit)

State machine

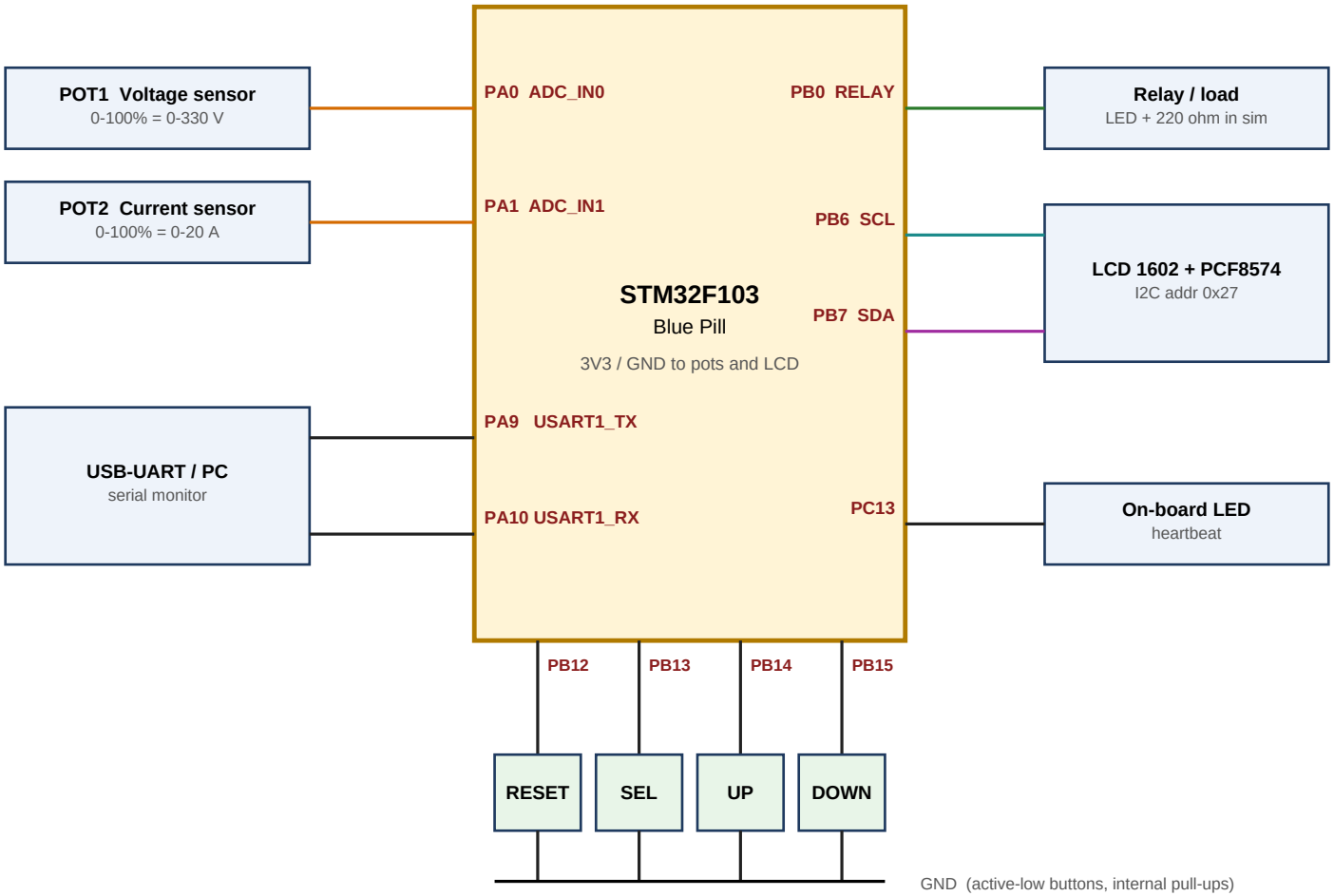
MONITORING -> (fault) DISCONNECTED -> COOLDOWN (3 s) -> MONITORING. Manual reset button on PB12 or the RESET command.

Changing settings

- **On device:** SEL cycles view / OV / UV / OC; UP/DOWN change (5 V or 0.5 A steps); 16x2 LCD; back to view after 10 s idle.
- **Serial** (115200 8N1): SET_OV:<V>, SET_UV:<V>, SET_OC:<A>, STATUS, RESET, HELP. OV > UV, OV <= 330 V, 0 < OC <= 20 A.
- Status line printed every second; PC13 heartbeat blinks 1 s normal, 200 ms on fault.

Limitations

Polled (not interrupt-driven) checks; settings in RAM only, lost on restart; no hardware watchdog.



Simulation vs. real product

Wokwi (project_files/diagram.json): pots stand in for the sensors, the LED stands in for the relay (lit = connected).
Real hardware still needs: mains voltage divider / transformer and current sensor (outputs within 0-3.3 V),
relay driver with flyback diode, and 4.7 k pull-ups on SCL/SDA if the LCD backpack has none.

Source: src/Core/Inc/config.h, project_files/diagram.json, readme.md